Piezoelectric Quantum 1/f Noise in AlGaN HFETs and Reliability

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ABSTRACT

The QED quantum 1/f noise formulas have been recently refined for the case of AlGaN/GaN HFETs and FETs through a better definition of the coherence parameter s, with much better agreement with the experiment. Indeed, for a FET/HFET width w>>L>t, this yielded $s \approx \pi nrtL\log(w/2L)$ instead of the old s=2nrtw formula. Here we generalize this basic result for the first time to a finite piezoelectric case. Here L= source to drain length, t is the thickness (depth) of the channel, n is the concentration of carriers, π=3.1416, and $r=2.8 \cdot 10^{-13}$ cm is the classical radius of the electron. In piezoelectric materials, particularly those also showing ferroelectric spontaneous polarization, transversal phonons are the massless quanta leading to large piezoelectric, or lattice-dynamic, quantum 1/f effects, conventional and coherent. As in the usual QED case, the parameter s' yields the observed 1/f noise as a weighted sum of conventional and coherent quantum 1/f effects. The HFET piezocoherence weighting parameter s', derived here is $(gN'\hbar/m^*v_s)(v_s/u)^3\ F(u/v_s)t/12w$, with $N'=nLt$, v_s the piezophonon speed, u the drift velocity, and F(x) is a function defined earlier, equal to $(2/3)x^3$ for x<<1. This s' is increasing, $\sim t^2$, important for reliability and device optimization. For HFET stability, a slower decrease of conductivity than of polarization is found to be needed for stability along the large device width, when the temperature increases.

Keywords: Piezoelectric Quantum 1/f noise, AlGaN/GaN HFETs optimiztion and reliability, Noise in HFETs and MODFETs, 1/f noise-based FET/HFET reliability optimization and testing.

1. INTRODUCTION

There are as many types of quantum 1/f noise [1]-[5], as there are systems of massless infraquanta with infrared-divergent coupling to the current carriers. Each of these types has a conventional and coherent part, corresponding to terms in the hamiltonian that are proportional with the first and second power of the number of carriers defining the current, respectively. In piezoelectric materials, particularly those also showing ferroelectric spontaneous polarization, transversal phonons are the massless quanta leading to piezoelectric, or lattice-dynamic, quantum 1/f effects, again both conventional and coherent. As in the usual QED case, the observable 1/f noise is approximated by a weighted sum of the conventional and coherent quantum 1/f effects. The sum involves the coherence parameter, this time denoted by s', in the weight factors. The parameter s' is also estimated from first principles. The general formula involving s' is applied to bulk GaN and AlGaN and thin sheets, and includes a dependence on geometry.

The best known form of quantum 1/f noise [1]-[2], [4] is certainly the electromagnetic, or QED form, that has been used to improve semiconductor devices other than the MOSFET and many other high-tech devices, sensors, and systems since the 1980s. Both the coherent and conventional QED-type Quantum 1/f Effect (Q1/fE) contribute to quantum 1/f noise, and are caused by the infrared divergent coupling between charged particles and soft photons. In final analysis, this new aspect of quantum mechanics is therefore due to the non-linearity of the system of particle and field, where the field reacts on the particles that generated the field in the first place.

The conventional Q1/fE is a new property of the "physical cross sections and process rates" introduced by one of us in order to allow quantum mechanics to explain fundamental 1/f noise in materials, devices and systems, including phase noise. It does not depend on the presence of other similar charged particles drifting in the electrical field. The coherent Q1/fE, on the other hand, is a basic property of any particle. It is caused by the uncertainty of its rest mass, due to the same infrared divergent coupling and due to the coherent state in which the photon states happen to be. This coherent state of the field is simply due to the presence of all the particles and to their motion.

The energy of coherent states is uncertain, as is known since Schrödinger introduced them in the 1920s. This uncertainty, derived from Heisenberg's uncertainty principle, causes the state of any particle to be non-stationary and beset by the fundamental coherent quantum 1/f noise $2\alpha/\pi f$, where $\alpha=e^2/\hbar c=1/137$ is Sommerfeld's fine structure constant. It is remarkable that the power spectral density of fractional fluctuations $2\alpha/\pi f$ in current, votage, mobility, diffusion constant, recombination speed, quartz resonator dissipation rate and frequency, etc., etc, contains Planck's constant in the denominator.

That makes this new aspect of quantum mechanics, or most important and fundamental form of quantum chaos, particularly interesting. It is the one that shapes our very existence, and causes us to like spontaneously, for instance, 1/f, or symphonic, music. Epistemologically, quantum 1/f noise is just a special case of the fundamental universal 1/f spectrum, caused by the simultaneous presence of nonlinearity and a special form of homogeneity in any system, e.g., traffic on the highways. This is described by Handel's "Sufficient Criterion" [1], that is satisfied also by QED at low frequencies. The drift motion of the current carriers, mentioned in the previous paragraph, causes a magnetic field and a magnetic form of kinetic energy, $LI^2/2$ where L is the inductance. In a thick copper wire, it is much larger than the mechanical form of the kinetic energy of the same drift motion of the current carriers. This magnetic energy of the drift motion of an electron is proportional to the number of other electrons, per unit length of the wire, that are drifting along with drift velocity v that is their average velocity. On the other extreme of a nanoscopically thin semiconductor, the mechanical form $mv^2/2$ dominates, as was pointed out already at the start of the theory of fundamental 1/f noise. Therefore we understand why the coherent Q1/fE is found to be dominant in a copper wire, while the conventional Q1/fE is found to be dominant in the very thin, poorly doped semiconductor whisker.

In Sec. 2 below we review the elementary relation between coherent and conventional Q1/fE [2] that is suggested by the relation between the magnetic and kinetic drift energies present in the Hamiltonian for the electromagnetic or quantum electrodynamical (QED) case, in which photons are the infraparticles. In Sec. 3 we repeat this more formally. In Sec. 4 this relation is derived again for the special case of FETs or HFETs, that have a width exceeding their length and thickness by orders of magnitude. In Sec. 5 we extend this treatment to the case of piezoelectric quantum 1/f noise, in which soft (transversal) piezophonons are the infraparticles. This was also introduced as Quantum Lattice-Dynamical (QLD) quantum 1/f noise.

The application to QLD allows us for the first time to explain the giant quantum 1/f noise in large ferroelectric semiconductor samples of BaSrTiO3, GaN and AlGaN and its startling decrease for smaller sample thickness. The speed of sound replaces the speed of light in the denominator of α, and the observed 1/f noise is 10^6 times larger, as expected from the theory. But there are no low frequency piezophonons, even in the largest semiconductor samples. In Sec. 6 we calculate the s' parameter for piezoelectric Q1/f noise in FETs and HFETs for the first time. Secs. 7 and 8 bring the experimental review of QLD Q1/f noise. Finally, our new HFET Stability & Reliability criterion is derived in Sec. 9.

2. ELEMENTARY DERIVATION OF THE s PARAMETER

We suggest that the relation of the coherent Q1/fE to the conventional Q1/fE is described in a first, rough, approximation through the parameter s defined here. The coherent state in a conductor or semiconductor sample is the result of the experimental efforts directed towards establishing a steady and constant current. It is, therefore, the state defined by the collective motion, i.e. by the drift of the current carriers. It is expressed in the hamiltonian by the magnetic energy E_m, per unit length of the current carried by the sample. In cylindrical samples or electronic devices, this coherent magnetic energy per unit length

$$E_m = \int(B^2/8\pi)d^3x = [nevS/c]^2[\mathit{ln}(R/r) + 1/4] \tag{2.1}$$

is much smaller than the total kinetic energy E_k of the drift motion of the individual carriers per unit length

$$E_k = \Sigma mv^2/2 = nSmv^2/2 = E_m/s. \tag{2.2}$$

Here we have introduced the magnetic field B, the carrier concentration n, the cross sectional area S and radius r of the cylindrical sample (e.g., a current carrying wire), the radius R>>r of the (e.g., circular) electric circuit, used as an upper cut-off for the integral in Eq. (2.1). One obtains the "coherence ratio"

$$s = E_m/E_k = 2ne^2S/mc^2[\ln(R/r)+1/4] \approx 2e^2N'/mc^2,= 2N'r_o. \tag{2.3}$$

Here N' = nS is the number of carriers per unit length of the sample, $r_o= e^2/mc^2=2.8\cdot10^{-13}$ cm is known as the classical radius of the electron, and the natural logarithm ln(R/r) has been approximated by one in the last form. We expect the total observed spectral density of the mobility fluctuations to be given by a relation of the form

$$(1/\mu^2)S_\mu(f) = [1/(1+s)][2\alpha A/fN] + [s/(1+s)][2\alpha/\pi fN] =[2\alpha/(1+s)fN][A+s/\pi], \tag{2.4}$$

which can be interpreted as an expression of the effective Hooge constant if the number N of carriers in the (homogeneous) sample is brought to the numerator of the left hand side. In this equation $\alpha A=2\alpha(\Delta\mathbf{v}/c)^2/3\pi$ is the usual nonrelativistic expression of the infrared exponent, present in the familiar form of the conventional quantum 1/f effect. This equation is limited to quantum 1/f mobility (or diffusion) fluctuations, and does not include the quantum 1/f noise in the surface and bulk recombination cross sections in the surface and bulk trapping centers, in tunneling and injection processes, in emission or in transitions between two solids.

Eq. (2.1) agrees with the well-known solution

$$L=(2/c^2)\{\mu[ln(8R/r)-2]+\mu'/4\} \tag{2.5}$$

for the inductance of a circular wire of radius R and thickness 2r in a medium of magnetic permeability μ, when the material of the wire has permeability μ'. The term $\mu'/4$ reflects the magnetic energy inside the wire, often neglected.

Note that the coherence ratio s introduced here equals the unity for the critical value $N' = N'' = 2{\cdot}10^{12}$/cm., e.g. for a cross section $S = 2{\cdot}10^{-4}$ cm^2 of the sample when $n = 10^{16}$. For small samples with $N' << N''$ only the first term survives, while for $N' > N''$ the second term in Eq. (2.4) is dominant. In practice, s is therefore twice the number of carriers in a salami slice perpendicular to the direction of the current, with a thickness given by the classical radius of the electron.

3. QED DERIVATION OF THE PARAMETER s FOR THE CYLINDRICAL CASE

The rationalized Gaussian system of units is used in this section that appears repetitive, but prepares the reader for the formulation of the piezoelectric case. In QED, the displacement amplitude of the electromagnetic oscillator with wave vector q and polarization ε is the Fourier component of the vector potential expanded over a cubic box of volume V

$$Z_q = \frac{ek^{\mu}\varepsilon_{\mu}}{\vec{k}\cdot\vec{q} - m\omega_q/\hbar}\cdot\frac{1}{\sqrt{2\hbar cqV}} \approx \frac{ek^{\mu}\varepsilon_{\mu}}{-mc}\cdot\frac{\hbar}{\sqrt{2\hbar cq^3V}} \tag{3.1}$$

It is produced by the presence of an electron of wave vector k, charge e and mass m. It is well known that the electron moving with velocity **v** generates a vector potential

$$A = \frac{e\vec{v}}{4\pi cr} = \frac{\vec{v}}{c}\Phi \Longrightarrow \xrightarrow{Fourier} \frac{e\vec{v}}{cq^2} = \frac{e\hbar\vec{k}}{mcq^2} = A_q \tag{3.2}$$

On the right we have shown the Fourier transform, and Φ is the electric potential. This coincides with Eq. (3.1), if we apply the normalization for Fourier expansion in a box of volume V. The current density of the electron is

$$\vec{j} = e\vec{v}\,\delta(\vec{r}) \xrightarrow{Fourier} e\vec{v} \tag{3.3}$$

The motional part of the electron's energy is

$$W_{motion} = \tfrac{1}{2}\int\frac{\vec{j}}{c}\cdot\vec{A}\,d^3r = \tfrac{1}{2}\frac{\vec{j}}{c}*\vec{A} = \tfrac{1}{2}\int e\vec{v}\,\frac{e\vec{v}}{cq^2}4\pi\frac{q^2dq}{(2\pi)^3} = \frac{e^2v^2}{4\pi^2c^2}q_{\max} = \frac{e^2v^2}{4\pi^2c^2}\frac{\pi}{2r_0} \tag{3.4}$$

Here * indicates a convolution. The parseval theorem was used and r_o is approximated by the classical radius of the electron $e^2/4\pi mc^2$, introduced in the previous section. The upper cutoff q_{max} introduced here for q, corresponds to a lower cutoff r_o for the first integral in Eq. (3.4). By carrying out that first integral we get

$$W_{motion} = \frac{e^2v^2}{8\pi r_oc^2}, \tag{3.5}$$

as indicated already at the end of Eq. 3.5, and by comparing with the q-space integral, we obtain $q_{max} = \pi/2r_o$.

In cylindrical samples or electronic devices, the coherent motional energy is the magnetostatic energy, dominated by the magnetic interaction energy between $N'=nS$ electrons per unit length in the direction of the current, with n being the electron concentration and S the cross section. Replacing $1/2r_o$ with $1/2r_o'=N'$ for each electron and multiplying with the number of electrons per unit length N', we obtain for the coherent magnetostatic energy per unit length

$$W_{coher} = \frac{e^2v^2N'}{8\pi r_0'c^2} = \frac{(envS)^2}{4\pi c^2} = \frac{(evN')^2}{4\pi c^2} \tag{3.6}$$

This allows us to reproduce the results in Eqs. (2.1) –(2.3) above in the rationalized Gauss System, often used in field theory, by multiplication with the well-known factor ln(R/r)+ ¼ and division with the KE/unit length $mN'v^2/2$

$$s = E_m/E_k = 2ne^2S/4\pi mc^2[\ln(R/r)+1/4] \approx 2e^2N'/4\pi mc^2, = 2N'r_o. \tag{3.7}$$

Here we just approximated the logarithmic factor again with unity in the last two forms. We thus regain Eq. (2.4).

4. DERIVATION OF THE PARAMETER s FOR THE FET and HFET CASES

A large value of the coherence parameter s should be avoided in electronic devices, because it favors the coherent 1/f noise component, which is much larger than the conventional one, due to the absence of the small factor $(v/c)^2$. We show here that a special geometry allows for a coherence parameter s that remains practically unaffected when the channel width is increased by several orders of magnitude over the channel length.

Modern FETs and HFETs attempt to increase the denominator N in the quantum 1/f formulas without markedly increasing the coherence parameter s, by choosing an extreme case of a rectangular cross section for the portion between the source and drain, and even more extreme for the channel, which is the region under the gate. Indeed, the circular cross section expected for a wire is replaced in this case by a short rectangle of very small length L_{DS}, with a very large width w of several hundreds of microns and less than a micron in thickness, the channel being even thinner. The channel is reduced to a 2-dimensional quantum sheet, which is a quantum well of size (thickness) t in the 3rd dimension, towards the gate. Most important, the "length" L_{DS} of the source-to-drain distance is only of the order of 1micron, while the gate length is just about a quarter micron or less. In this case the kinetic energy E_k per unit length in the direction of L_{DS} is still given by $mv^2/2$, where v is the drift velocity, but the magnetic energy E_m per unit length in the direction of L_{DS} is roughly proportional with the first power of w only, instead of w^2, and can be approximated by

$$E_m = \pi[\ln(w/2L_{DS})]L_{DS}[nevS/c]^2/w. \tag{4.1}$$

This indicates decoherence along the device width w. For $w>>L_{DS}>t$, this yields a coherence ratio

$$s \approx \pi n r_o t L_{DS} \ln(w/2L_{DS}) \qquad \text{for } w>>L_{DS}>t, \tag{4.2}$$

which shows that only an effective width $w=w_{eff}$ about equal to L_{DS} should be used in the calculation of the coherence parameter s and of the in this special case; larger widths are subject to de-coherence, i.e., their magnetic energy no longer increases proportional to w^2, or N'^2. This favors lower, mainly conventional, quantum 1/f noise in these devices, in spite of the large values of w, in agreement with the practical results, which anticipated the quantum 1/f result simply by trial and error, unaware of the subtleties of the transition from coherent to conventional Q1/f noise in this case, and in general. It also shows that fragmentation of the extremely large widths w does not reduce 1/f noise in this special case.

To derive the rough approximation given in Eq. (4.2), one writes the expression of the coherent ($\propto n^2$) contribution of the currents in the short, but very wide channel to the magnetic energy of the circuit in the form

$$E_m=\int[\mathbf{j}(\mathbf{x})\cdot\mathbf{j}(\mathbf{x'})/2c^2|\mathbf{x}-\mathbf{x'}|]d^3x d^3x' = e^2 v_d^2 \int[n(\mathbf{x})\cdot n(\mathbf{x'})/2c^2|\mathbf{x}-\mathbf{x'}|]d^3x d^3x'. \tag{4.3}$$

The integral coincides with the electrostatic energy of an equivalent constant charge density numerically equal to e/c^2 in CGS, or $e\mu_o/4\pi\varepsilon_0$ in SI, uniformly distributed over the volume wtL_{DS} of the channel. This energy, in turn, is calculated as the corresponding volume integral over the squared equivalent electric field, using L_{DS} as a lower cut-off. The equivalent electric field is $E=netw/cr(r^2+w^2/4)^{1/2}$ at a distance r in the median plane of a segment of length w oriented along (aligned with) the width w of the channel. In performing the equivalent integral over $E^2/8\pi$ over the whole space, we use this to approximate the field in space outside the length and along the width w of the channel, because w is very large.. A more elaborate calculation does not simply approximate the field with its value in this median plane, but yields similar results in the end, as Eq. (4.1).

In conclusion, Eq. (4.1) just replaces the factor ln(R/r) of Eq. (2.1) according to the rule

$$\ln(R/r) \longrightarrow \pi[\ln(w/2L_{DS})]L_{DS}/w. \tag{4.4}$$

We now have all the ingredients for a discussion of the piezoelectric, i.e., Quantum Lattice-Dynamic (QLD) Quantum 1/f Effect (Q1/fE) case.

5. CONNECTION OF COHERENT AND CONVENTIONAL QLD-Q1/fE

The conventional, or incoherent, piezoelectric quantum 1/*f* noise effect [3], is given by

$$\frac{S_\sigma(f)}{\langle\sigma\rangle^2} = \frac{2gA}{fN}, \tag{5.1}$$

and is dominant in small samples or devices, where $A = 2(\Delta v)^2/3\pi(v_s)^2$, Δv is the change of electron velocity during scattering and σ is any scattering cross section defined for carriers that exhibit piezoelectric coupling to ELF phonons. Here g is the dimensionless piezopolaron coupling constant, similar to the fine structure constant α, but with the speed of transversal sound waves replacing the speed of light in the denominator. It also involves a geometrical factor of the order of unity, dependent on the lattice. In a first approximation, σ can also be the mobility μ, the conductivity, or the current *j*. In the *coherent case* [4] of large devices,

$$\frac{S_j(f)}{j^2} = \frac{4g}{\pi f}\frac{1}{\left[1-(\hbar k_s/m^* v_s)^2\right]}. \tag{5.2}$$

Here $(\hbar k_s/m^*)$= |**v**| = u is the drift velocity of the carriers, m* their effective mass, and v_s is the speed of sound for transversal acoustic phonons.

We now derive the connection between the coherent and conventional forms of Lattice-Dynamical (piezoelectric) Quantum 1/f Effect (LD-Q1/fE) similarly to the way this was done in 1985 for the usual QED-Q1/fE [3].

In QLD, the amplitude of the piezo-elastic transversal lattice mode with wave vector **q** is the Fourier component **q** of the displacement **B** caused by an electron in the lattice, expanded over a cubic box of volume V

$$B_{\vec q} = \frac{-iv_s(2\pi g/q^3V)^{1/2}}{v_s - \hbar\hat q\cdot\vec k/m} \tag{5.3}$$

This is similar to Eq. (3.1). The part corresponding to the motion is obtained by subtracting the static part (Eq. 5.3 for k=0) that applies for an electron at rest

$$B_{\vec q}^{mot} = -iv_s\left(\frac{2\pi g}{q^3V}\right)^{\frac12}\left[\frac{1}{v_s-\hbar\hat q\cdot\vec k/m}-\frac{1}{v_s}\right] = -iv_s\left(\frac{2\pi g}{q^3V}\right)^{\frac12}\frac{\hbar\hat q\cdot\vec k}{mv_s(v_s-\hbar\hat q\cdot\vec k/m)} \tag{5.4}$$

The corresponding motional energy W^{mot} for an electron of wave vector k is thus

$$v_s^{-2}W_{\vec k}^{mot} = \sum_{\vec q}|B_{\vec q}^{mot}|^2\,\hbar\omega_q = \frac{2\pi g}{V}\frac{\hbar^3}{m^2v_s}\frac{V}{(2\pi)^3}\int d\Omega\int_0^{q_D}dq\frac{k^2\cos^2\theta}{(v_s-\hbar k\cos\theta/m)^2} \tag{5.5}$$

Here q_D is the Debye wave vector. This is similar to Eq. (3.4), but here the second term in the denominator was not neglected, since v_s<< c. The integration yields

$$W_{\vec k}^{mot} = \frac{g\hbar^3}{2\pi m^2v_s}\frac{V}{(2\pi)^3}\int_0^{q_D}dq\int_{-1}^{1}\frac{k^2x^2dx}{(1-\hbar kx/mv_s)^2} = \frac{g}{2\pi}\frac{mv_s^2q_D}{k}F(\frac{v}{v_s}) \tag{5.6}$$

With q_D=1/r'$_0$, we obtain thus for cylindrical symmetry

$$W_{\vec k}^{mot} = \frac{g}{2\pi}\frac{mv_s^2}{kr_o'}F(\frac{v}{v_s})\xrightarrow{Cylindrical-Symm}\frac{g}{2\pi}\frac{mv_s^2}{kr_o'}F(\frac{v}{v_s})N'[\ln\frac{R}{r_0'}+\frac14] \tag{5.7}$$

N' is again the number of carriers per unit length of the sample in the direction of current flow, and *m** is the electron effective mass. The integration in Eq. (5.6) introduced the function

$$F(u/v_s) = 2(u/v_s)\{1 + 1/[1 - (u/v_s)^2]\} + \ln[1-(u/v_s)]^2 - \ln[1+(u/v_s)]^2. \tag{5.8}$$

We introduce the ratio of the coherent piezoelectric energy of the drift motion divided by the additional kinetic energy of the carriers introduced by the drift velocity u.

$$s' = \frac{\frac{g}{2\pi}\frac{mv_s^2}{kr_o'}F(\frac{u}{v_s})N'\ln\frac{R}{r_0'}}{N©mu^2/2} = \frac{g}{\pi}\frac{v_s^2}{u^2kr_o'}F(\frac{u}{v_s})[\ln\frac{R}{r_0'}+\frac14] = (gN'\hbar/\pi m^*v_s)(v_s/u)^3F(u/v_s)[\ln(R/r_o')+1/4]. \tag{5.9}$$

For an infinite piezoelectric medium, with a cylindrical current-carrying portion, this is the desired ratio s' of terms present in the hamiltonian, corresponding in QLD to the parameter s. If only the cylindrical sample is piezoelectric we drop the logarithm and keep only the term ¼, since this would correspond to having only the term with μ' in Eq. (2.5):

$$s' = (gN'\hbar/\pi m^*v_s)(v_s/u)^3F(u/v_s)[\ln(R/r_o')+1/4]. \tag{5.10}$$

As before, the coherent and conventional forms of QLD-Q1/fE are superposed with weights s/(1+s) and 1/(1+s). For the intermediate case, i.e., for sizes in between coherent and incoherent, we have $S_j(f)/j^2 = \alpha/Nf$, with α defined as

$$\alpha = \alpha_{incoh}\frac{1}{(1+s')}+\alpha_{coh}\frac{s'}{(1+s')} \tag{5.11}$$

Again, similar to s, the weight s' is the ratio between the piezoelectric or LD form of kinetic energy [associated with the electric current present in the sample or device, that is coherent], and the incoherent kinetic energy $m_{eff}u^2/2$ of individual particles, [caused by the drift velocity u].

The function F present in the definition of s' is given by Eq. (5.8). For u/v_s << 1, $F(u/v_s) = (2/3)(u/v_s)^3$; then *s*' can be simplified as *s*' = $2gN'\hbar/3\pi m^*v_s$. This expression can be used for small drift velocity, much smaller than the sound velocity in the semiconductor. For $1-(u/v_s) << 1$, $F(u/v_s) = 1/[1-(u/v_s)]$, then s' can be approximated as $s' = (gN'\hbar/\pi m^* v_s)(v_s/u)^3\{1/[1-(u/v_s)]\}$, which is very large. This expression is used for drift velocities approaching the speed of sound.

6. THE PARAMETER s' FOR THE FET and HFET CASES

Considering again the extreme geometry of very large width $w>>L_{DS}>t$ discussed for FETs and HFETs in Sec. 4, we realize that the mathematical equivalence of the magnetic and piezoelectric attractive interactions between current carriers tempts us to repeat the steps that led to Eq. (4.1) above. The vector potential $\mathbf{A_q}$ of Eq. (3.4) is just replaced by the displacement amplitude $\mathbf{B_q}^{mot}$ of Eq. (5.4). This would suggest, in analogy to Eq. (5.7), to try an expression of the type

$W_k^{mot}=(g/2\pi)(mv_s^2/kr_o')F(v/v_s)N'\pi[\ln(w/2L_{DS})]L_{DS}/w$ .... wrong!

in which the logarithm was replaced as shown in (4.4). However, there is a major difference here, causing this expression to be wrong. The magnetic field extends over all of space, while the piezoelectric stress is present only in the sample. Therefore, the integral over the squared electric field that is equivalent to Eq. (4.3) in the expression of the magnetic energy E_m must be limited to the space inside the sample in this case, in order to model the piezoelectric system. Applying the Gauss theorem with a constant concentration of carriers n(x)=const of charge e, we obtain an electric field 4πnet'Lw/Lw, with t' integrated from (–t/2) to t/2, and an equivalent magnetic energy

$$E_m'=(e^2v_d^2/2c^2)\int[n(\mathbf{x})n(\mathbf{x}')/|\mathbf{x}-\mathbf{x}'|]=(v_d^2/8\pi c^2)\int E^2d^3x$$
$$=(n^2e^2v_d^2/2c^2)4\pi Lw\int t'^2dt'=(n^2e^2v_d^2/2c^2)\pi Lwt^3/6=\pi(ev_dN'/c)^2Lt/12w. \qquad (6.1)$$

In conclusion, Eq. (6.1) just replaces the factor ln(R/r) of Eq. (2.1) according to the rule

$$\ln(R/r) \longrightarrow \pi t/12w. \qquad (6.2)$$

This yields the final result

$$W_k^{mot}=(g/2\pi)(mv_s^2/kr_o')F(v/v_s)N'\pi t/12w = (gN'^2\hbar/\pi)(v_s^2/u)\,F(u/v_s)\,\pi t/12w. \qquad (6.3)$$

Finally, we obtain this way the Quantum Lattice Dynamic coherence parameter s' in the form

$$s' == (gN'\hbar/m^*v_s)(\,v_s/u)^3F(u/v_s)t/12w. \qquad (6.4)$$

This is the sought expression of the piezoelectric coherence parameter s' for the special case of FETs and HFETs with w>>LDS>t derived here for the first time. It can easily be generalized to the case of several sheets with different piezoelectric properties in the above integral over t'. We notice that the coherent contribution is strongly reduced due to the smallness of t, as compared to w.

7. EXPERIMENTAL TEMPERATURE-DEPENDENT RESISTIVITY

The resistivity of both ELO-GaN (epitaxial lateral overgrowth generated) and n^+-GaN:Si (doped) samples was measured [5] for temperature varying from 100-550K. As illustrated in Fig. 7.1, resistivity decreases as temperature increases (100-300K) for both ELO-GaN and n^+-GaN:Si. This is attributed to the fact that ionized impurity scattering is dominating [6]. As the temperature increases further (300-530K), the resistivity for ELO-GaN decreases, but that for n^+-GaN:Si increases. The later phenomenon is attributed to crystal defect scattering [6]. Since the presence of a lattice defect is always associated with an increase of the electrical resistivity and the fact that the dislocation density of ELO-GaN is significantly lower compared to n^+-GaN:Si, hence the resistivity of ELO-GaN decreases with increasing temperature. Fig. 7.1 also reveals that the change of the measured resitivity for ELO-GaN at high temperature (300–500K) is negative whereas for n^+-Ga:Si is positive. The resistance change is expressed as [7]

$$\frac{\Delta R}{R} = \frac{\Delta\rho}{\rho} + 4\gamma\frac{1-2\nu}{1-\nu}s|\Delta\varepsilon| = \frac{\Delta\rho}{\rho} + s\beta|\Delta\varepsilon| \qquad (7.1)$$

where $\Delta\rho = \rho_D\Delta n_D$; Δn_D is the variation of dislocation density; ρ_D is the dislocation resistivity; γ is the Grüneisen's constant; ν is the Poisson ratio; Δε is the strain variation and $s = \pm 1$ according to whether the stress is compressive (s = +1) or tensile (s = -1). The total resistivity ρ contains vacancies, interstitials and dislocations with respective densities n_v, n_I and n_D. It is usually expressed as $\rho = \rho_{ph} + \rho_v n_v + \rho_I n_I + \rho_D n_D$ where ρ_{ph} represents the phonon contribution. The term $s\beta|\Delta\varepsilon|$ incorporates the piezoelectric resistance changes. Since the thermal expansion coefficients of GaN and Al_2O_3 are respectively $\alpha_{GaN} = 5.45\times10^{-6}/°C$ and $\alpha_s = 7.5\times10^{-6}/°C$ [8], and $\alpha_{GaN} < \alpha_s$, heating from the unstressed state leads to tension, cooling to compression. For GaN material, γ = 1.8 [9], therefore, the term incorporating the piezoelectric resistance changes $s\beta|\Delta\varepsilon|$ in (7.1) is negative since s is negative (tension) at high temperature. From Fig. 7.1, we can infer that $s\beta|\Delta\varepsilon| > \Delta\rho/\rho$ for ELO-GaN in Eq. (22) and the reverse is true for n^+-GaN:Si. This leads us to conclude that dislocation rearrangement (and other defects) dominates in n^+-GaN:Si whereas piezoelectric effect dominates for ELO-GaN at high temperature.

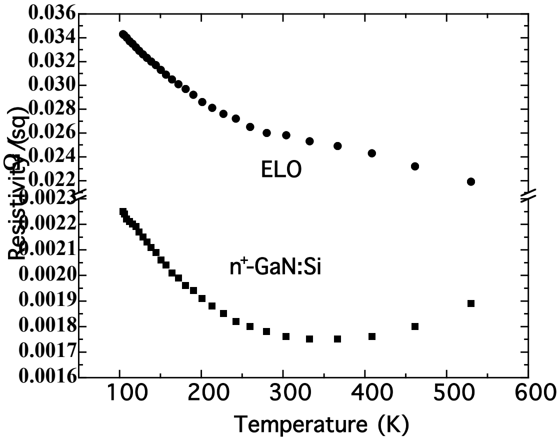

Fig. 7.1: Temperature-dependent resistivity of ELO-GaN sheets and n^+-GaN:Si in Ω/square [5].

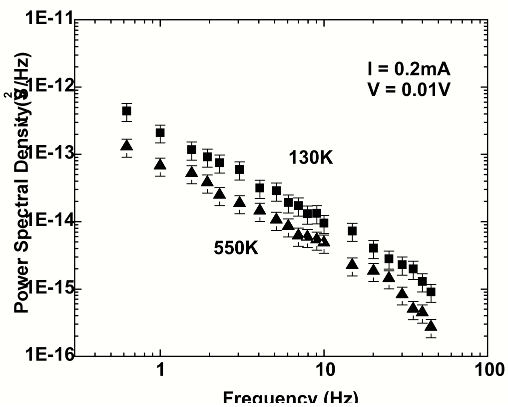

Fig. 8.1: Power spectral densities of voltage fluctuations S_V, for ELO-GaN at 550K. The S_v at 130K is included for comparison, V^2/Hz.

8. LOW FREQUENCY NOISE CHARACTERISTICS

In evaluating the noise parameter α, the operating voltage range was selected to be less than 0.05V. The voltage power spectral density S_V with error bars is shown [5] in Fig. 8.1, for ELO-GaN measured at 130K and 550K at a constant biasing current of 0.2mA. The spectra show $1/f$ behavior below 50Hz. The noise parameter can be estimated experimentally as $\alpha = S_V fN/V^2$ [10]. At 550°C, the measured carrier concentration for ELO-GaN is 3.2×10^{16}cm^{-3} and carrier mobility is 132cm^2/Vs. The total carrier number within the 8μm thick bridge-shaped Hall-bar ELO-GaN is $N = 2.04\times10^8$. With a measured voltage of 0.01V, $\alpha \approx 0.66$ was obtained. The measurement errors of S_V at 550K for selected 20 frequency points within the range of 0.1-50Hz are estimated to be about 10% [5].

Fig. 8.1 illustrates only the voltage power spectral densities S_V of ELO-GaN at 130K and 550K. For temperatures within this range, the noise parameters α were also evaluated and shown in Table 8.2. Note that the intercept points were taken with the gradient fixed at –1. These experimentally determined noise parameters α were also plotted [5] against temperature variation as shown in Fig. 8.2. At high and low temperatures, the noise parameter α tends towards higher values, compared to α at room temperature. The possible explanation for this temperature-dependent behavior of α may

Temperature (K)	Intercept	α
130	-12.106	1.6
150	-12.094	1.64
200	-12.410	0.79
250	-12.504	0.639
300	-12.473	0.686
350	-12.624	0.485
400	-12.586	0.529
450	-12.598	0.515
500	-12.575	0.543
550	-12.490	0.66

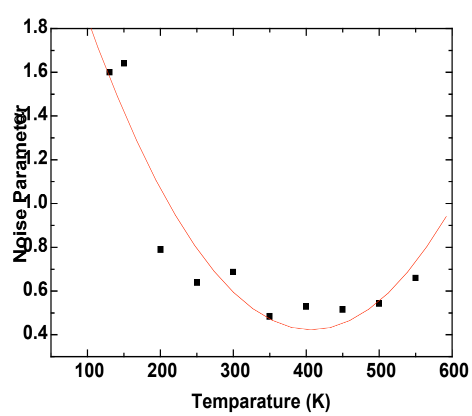

Fig. 8.2: Temperature dependence noise parameters α for ELO-GaN [5].

Table 8.2: Temperature dependence noise parameters α for ELO-GaN [5].

be due to the possible increase of the piezopolaron coupling coefficient g. An investigation into this strain-dependent g will be carried out in the subsequent section [5].

8.1_The Macroscopic Piezoelectric Equations

The *macroscopic piezoelectric equations* that relate the stress T_α, strain S_β, electric fields E_i and displacement field D_i are expressed as

$$T_\alpha = c_{\alpha\beta}S_\beta - e_{k\alpha}E_k \tag{8.1}$$

$$D_k = 4\pi e_{k\alpha}S_\alpha + \varepsilon_{ik}E_i \tag{8.2}$$

Note that repeated indices imply summation. The subscripts i, j, or k are (x,y,z) while α or β means xx, yy, zz, yz, xz, xy. The constant parameters are the familiar elastic constant $c_{\alpha\beta}$, the dielectric tensor ε_{ik}, and the piezoelectric constant $e_{k\alpha}$. Consider now the piezoelectric stiffening of the crystal, equation (8.1) can then be re-written as [14]

$$T_\alpha = \bar{c}_{\alpha\beta}S_\beta \tag{8.3}$$

where
$$\bar{c}_{\alpha\beta} = c_{\alpha\beta} + \frac{q_k e_{k\beta} q_i e_{i\alpha}}{q_m \varepsilon_{mn} q_n} \tag{8.4}$$

Notice the term $q_k e_{k\beta} q_i e_{i\alpha}/q_m \varepsilon_{mn} q_n c_{\alpha\beta}$, is actually the electromechanical coupling coefficient $\langle f \rangle$. Expression (4) shows that the piezoelectric interactions affect the elastic constants, and hence the speed of sound where $c = \rho v^2$. This in turn affects the piezopolaron coupling coefficient g, which is an important parameter for the determination of the noise parameter α to be discussed in the subsequent sections [5].

8.2_Piezopolaron Coupling Coefficient

Let us now consider more closely the dimensionless piezopolaron coupling coefficient g. For transversal acoustic phonon, we write the expression for g as [14]

$$g = \frac{e^2}{\hbar\varepsilon_0 \langle v_s \rangle}\langle f \rangle \tag{8.5}$$

As usual, we express $\langle f \rangle = q_k e_{k\beta} q_i e_{i\alpha}/q_m \varepsilon_{mn} q_n c_{\alpha\beta}$, which is the angular average electromechanical coupling coefficient in the transversal acoustic mode. It is a measure of the strength of the electron-phonon coupling and it varies over direction of phonons, therefore its value is usually obtained by taking some average over different phonon directions.

Consider now the macroscopic piezoelectric equation given in (1), if there is no *tensile* or *compressive strain*, i.e., $S_\beta = 0$, which is a typical case for free-standing GaN epilayer such as ELO-GaN, then the built-in stress will be $T_\alpha = - e_{k\alpha}E_k$. Here the electric field E_k is expressed as $E_k = 2e_{k\alpha}\Delta\varepsilon/\varepsilon_0\varepsilon$, where $\Delta\varepsilon = \varepsilon_\parallel - \varepsilon_\perp$, is the *tetragonal distortion* of the piezoelectric layer. Note that E_k is an average value and the averaging has been made by the angular average of the piezoelectric coupling constants $e_{k\alpha}$ written as [7],

$$\langle e_{ijk}^2 \rangle = \frac{2}{35}(e_{33} - e_{31} - e_{15})^2 + \frac{16}{105}e_{15}(e_{33} - e_{31} - e_{15}) + \frac{16}{35}e_{15}^2 \tag{8.6}$$

Wurtzite-structure GaN has three independent piezoelectric constants, e_{15}, e_{31}, and e_{33} with respective values of -0.30Cm^{-2}, -0.33Cm^{-2} and 0.65Cm^{-2}. The dielectric tensors are $\varepsilon_{11} = \varepsilon_\perp = 8.6$ and $\varepsilon_{33} = \varepsilon_\parallel = 10.1$. Therefore, E_k can be estimated to be $E_k = 6.281\times10^9$ V/m. Since we are concerned only with the shear mode along the c axis, the built-in shear stress can be evaluated as $T_{xx} = T_{yy} = 1.88$ GPa. This is a theoretical calculated shear stress, in a typical GaN epilayer, however, the built-in shear stress is in the order of < 1.5 GPa. The possible explanation for the difference between the theoretical and experimental shear stresses may be due to the formation of misfit dislocations, hence, typical GaN epilayer exhibits lower shear stress compared to the calculated value.

The piezoelectrically stiffened transversal acoustic phonon velocity is expressed as $v_s = \sqrt{c_{ii}\left(1 + k^2\right)/\rho}$. Here, c_{ii} is the elastic modulus relevant for the bulk acoustic-wave, $\rho = 6150$ kg/m^3 is the mass density of GaN and $k^2 = \langle f \rangle$ is the electromechanical coupling coefficient, which leads to piezoelectrically stiffened elastic constants if displacement along the c axis is involved in the wave propagation. For shear mode, $k^2 = e_{15}^2/\varepsilon_0\varepsilon_{11}c_{44}$. Substituting for k^2, $v_s = \sqrt{\left(c_{44} + e_{15}^2/\varepsilon_0\varepsilon_{11}\right)/\rho}$. In order to obtain a shear strain dependent transversal acoustic phonon velocity, we need to express c_{44} as $(T_{11}+e_{15}E_k)/S_{11}$. Therefore, we obtain [5]

$$v_s = \sqrt{\left[(T_{11} + e_{15}E_k)/S_{11} + e_{15}^2/\varepsilon_0\varepsilon_{11}\right]/\rho} \qquad (8.7)$$

Note that the second term of the above expression contributes less significantly to the transversal acoustic phonon velocity v_s, therefore, it is sufficient that only average values for the piezoelectric coupling constant $\langle e_{ijk}^2 \rangle$ and static dielectric constant ε_{11} will be used. The assumptions made here, however, are that the electric field $E_k = 6.281\times10^9$ V/m as well as the built-in shear stress T_{11} = 1.88 GPa are not changing with the shear strain. These assumptions are made in the belief that for any increment of shear strain, the changes in the shear stress T_{11} and electric field E_k will be accommodated by the formation of misfit dislocations. However, the point worth mentioning here is that there is a singularity occurring at $S_{11} = 0$. As a result, we need to limit the S_{11} value to -1×10^{-6}. The negative sign indicates compressive strain. Fig. 8.5 shows a theoretical evaluated shear strain dependent transversal acoustic phonon velocity. The plot [5] reveals that the velocity approaches a high value of about 7×10^3 m/s at $S_{11} = -1\times10^{-6}$. As the shear strain increases, i.e., it becomes more compressive, the velocity decreases to a constant minimum value of about 1×10^3 m/s at a strain value of around -1×10^{-5}.

Consider now the electromechanical coupling coefficient $\langle f \rangle = q_k e_{k\beta} q_i e_{i\alpha}/q_m \varepsilon_{mn} q_n c_{\alpha\beta}$. For transversal acoustic phonon, $\langle f \rangle$ can be expressed as $\langle f \rangle = k^2 = e_{15}^2/\varepsilon_0\varepsilon_{11}c_{44} = e_{15}^2/\rho\varepsilon_0\varepsilon_{11}v_s^2$ where $c_{44} = \rho v_s^2$. Since v_s is strain dependent as expressed in (8.7), this implies that $\langle f \rangle$ is also strain dependent. Nevertheless, similar to the situation of the electric field E_k and shear stress T_{11}, we have chosen to fix the electromechanical coupling coefficient $\langle f \rangle$ at zero shear strain and observe the strain dependent piezopolaron coupling coefficient g. From expression (8.5), a strain dependent g is plotted as shown in Fig. 8.3. It shows that g varies approximately from 1.36 at $S_{11} = -1\times10^{-6}$ to 8.85 at $S_{11} = -5\times10^{-5}$.

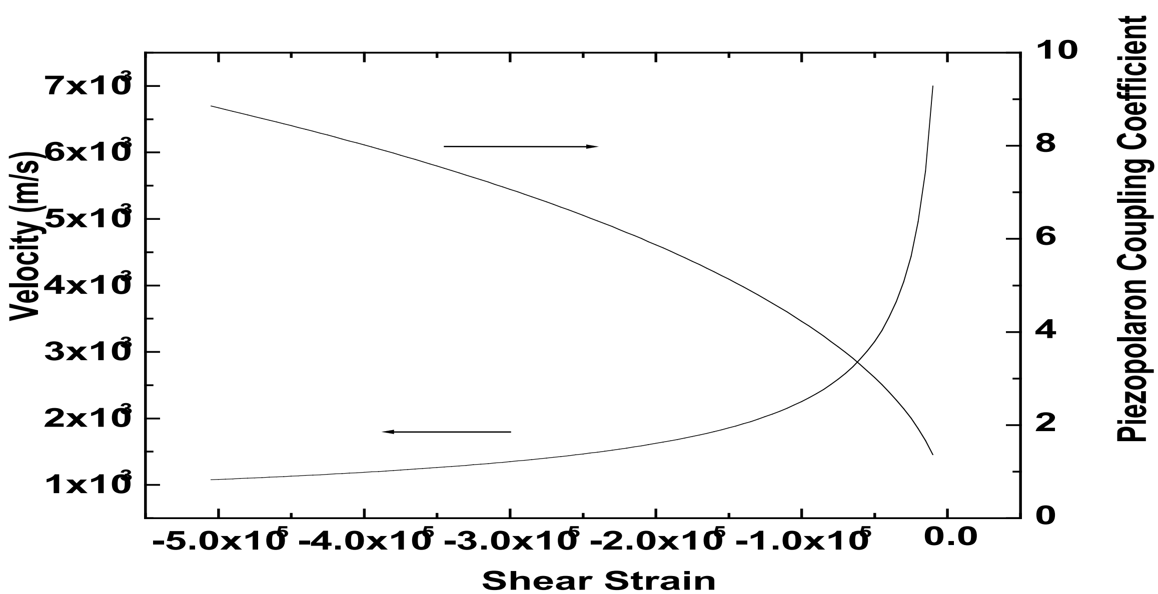

Fig. 8.3: Strain dependence of velocity of transversal acoustic phonon v_s and piezopolaron coupling coefficient g.

8.3_Low Frequency Noise in Epitaxial Lateral Overgrown GaN

In our ELO GaN, we evaluated the shear strain S_{11} by measuring the frequency shift of the high frequency E_2(TO) phonon utilizing Raman spectroscopy. The strain tensor components are given as $S_{11} = S_{22} = \Delta\omega_\lambda/(2a_\lambda+b_\lambda q)$ and $S_{33} = qS_{11}$. Here, λ is the E_2(TO)(High) phonon mode, $\Delta\omega$ is the change of frequency, a and b are the phonon deformation potential constants and $q = -2c_{13}/c_{33}$. In our ELO-GaN, the measured change of E_2(TO)(High) phonon frequency $\Delta\omega_{E2}$ was negligible. With $a = -850\text{cm}^{-1}$ and $b = -920\text{cm}^{-1}$ and using c_{13} = 106 GPa and c_{33} = 398 GPa, we estimated the shear strain tensors $S_{11} = S_{22} = -5\times10^{-6}$. This corresponds to a piezopolaron coupling coefficient of $g = 3$ [5].

At low electric field the drift velocity u is proportional to the electric field strength E, i.e., $u = \mu E$. For our ELO-GaN, the measured mobility was $\mu = 132 cm^2/Vs$ at 550K. With an applied voltage of 0.01V, the electric field evaluated was $E = 0.625 V/cm$, this corresponds to $u = 82.5 cm/s$. Since u is very much smaller than v_s, we employed the expression $s' = 2gN'\hbar/3\pi m^* v_s$ from Eq. (5.1) to evaluate s'. With $N' \sim 10^8$ carriers/cm length for ELO-GaN, s' evaluated is about 1000. Since s' is large, it is sufficient to use coherent piezoelectric quantum $1/f$ noise theory. Indeed, the large s' causes the weight $1/(1+s')$ of the conventional quantum $1/f$ term to be comparatively small, and the weight of the coherent term to be almost unity. As a result, the quantum $1/f$ coefficient α is simplified to $\alpha = 2g/\pi$ which yields 2 for $g = 3$. This result correlates well with our experimentally obtained α which has values within the range 0.5-2.

Note that slight non-uniformities present in any sample could slightly increase the theoretical α value above 2. They could also down-correct the measured α to values below 0.5, because the assumed number of carriers in the expression $\alpha = S_V fN/V^2$ used by us to define the measured α may have been slightly larger than the effective number that remains when non-uniformities are taken into account.

8.4 Low Frequency Noise in n^+-GaN:Si

For our n^+-GaN:Si sample, the measured change of E_2(TO)(High) phonon frequency was $\Delta\omega_{E2} = 0.6 cm^{-1}$. The estimated shear strain tensors were $S_{11} = S_{22} = -1\text{-}3\times10^{-5}$. The measured [5] carrier mobility for our n^+-GaN:Si sample was $\mu = 100 cm^2/Vs$ at room temperature. With an applied voltage of $V < 0.1V$, the electric field $E = 6.25 V/cm$, this corresponds to $u = 625 cm/s$. Similar to the case of ELO-GaN, with u small compared to v_s and since $N' \sim 10^9$ carriers/cm length for n^+-GaN:Si, it is sufficient to use the coherent piezoelectric quantum $1/f$ noise theory. As a result, the estimated quantum $1/f$ coefficient α was between 4 and 7. This result correlates well with our experimentally obtained α value, which is 5.

9. OPTIMIZATION AGAINST A NEW TYPE OF INSTABILITY ALONG THE HFET'S LARGE WIDTH, LIMITING RELIABILITY

Consider an HFET that is uniform in the direction of the large width of the device (e. g., width Z=200 μm, with a submicron length), with a uniform heat convection loss rate per unit width $r(T-T_o)$, where r is the heat exchange coefficient with the surroundings that are at a temperature T_o. The heat diffusion equation along the width coordinate z is

$$C\partial T/\partial t - \kappa\nabla^2 T = -r(T-T_o) + \sigma V^2 + \sigma_G V_G^2, \qquad (9.1)$$

where C is the heat capacity per unit width in the z direction, κ the heat conduction coefficient in the z direction, $\sigma(T, V_G)$ is the source to drain electrical conductance per unit width, $V=V_{DS}$ the applied source to drain voltage, $\sigma_G(T)$ the conductance per unit width of the gate insulation, and $V_G = V_{GS}$ the gate to source voltage. The unperturbed stationary form of Eq. (9.1) is

$$r(T-T_o) = \sigma(T, V_G)V^2 + \sigma_G(T)V_G^2, \qquad (9.2)$$

Consider now an arbitrarily small temperature perturbation. Without loss of generality, it can be considered as a superposition of Fourier components

$$T' = A\exp[ikz-\omega t], \qquad (9.3)$$

with arbitrarily small $A_{k,\omega}$. Substituting the solution T+T' into Eq. (9.1) and subtracting Eq. (9.2) we obtain the linearized equation

$$(-i\omega C + \kappa k^2)T' = -rT' + (\partial\sigma/\partial T)T'V^2 + (\partial\sigma_G/\partial T)T'V_G^2, \qquad (9.4)$$

This yields the heat stability criterion (H-criterion)

$$-i\omega C = -r - \kappa(\pi/L)^2 + pV^2 + qV_G^2 < 0. \qquad (9.5)$$

here we have set $k = n\pi/L$ with n=1. The modes with higher n will then also be stable, having a negative time-dependent exponent that exponentially attenuates the initial perturbation. If the parameters r, κ, p and q, averaged over the length from source to drain, satisfy this inequality (9.5), the HFET or FET is expected to be stable against temperature fluctuations leading which would otherwise lead to the filamentation instability, provided we neglect the self-constricting action [22] of the own magnetic field of the temperature and current fluctuation. This self-constriction [22] is particularly large in the presence of a p-n junction or n-n+ junction with reverse bias, but less important in the FET and HFET case with low gate currents. Since q>0, this limits the applied gate voltage in general, for which stability can be achieved. It depends very much on $p=p_1+p_2$, which has two components of opposite signs. Indeed,

$$p = p_1 + p_2, \text{ with } p_1 = \partial\sigma/\partial T_{th\ activ} = E_a\sigma/T^2 > 0, \qquad (9.6)$$

where E_a is the relevant thermal activation energy for carriers in the (doped or unintentionally doped) channel. We also introduced the usually negative

$$p_2 = \partial\sigma/\partial T_{polariz} = (\partial\sigma/\partial P)\ (\partial P/\partial T) < 0. \tag{9.7}$$

Indeed, the spontaneous polarization of AlGaN/GaN HFET gate insulation layers decreases with temperature. The polarization of high-ε FET insulation layers also decreases with temperature. For stability, the absolute magnitude of p_2 should be chosen large enough to satisfy the inequality (9.5), which also implies p<0.

In conclusion, to optimize a FET against the H-instability, we need to provide enough high-ε gate dielectric with a sufficiently high negative temperature derivative of ε, in order to satisfy the criteria (9.5) or

$$p = p_1 + p_2 < 0 \tag{9.8}$$

at the chosen operational value of the gate voltage V_G. For an HFET with spontaneously polarized gate insulation generating the current carriers in the channel, a sufficiently large negative temperature derivative of the magnitude of the spontaneous polarization is needed in order to satisfy conditions (9.5) and (9.8).

Experimentally, the high temperature operation of AlGaN/GaN HFETs with normal electrical bias, leads to damage of the channel that is consistent with what could be initiated by the H-instability introduced here. According to the studies of Sozza et al. [23]-[24], the high temperature operation (HTO) test, studying AlGaN/GaN HFETs at high temperature under normal bias conditions, proved to be by far the most damaging, leading to the highest degradation of the electrical parameters of the transistors. It also led to a more than 2 orders of magnitude increase of the normalized (fractional) 1/f noise power spectral density, as shown in Fig. 5a, 5b, and 8 of Sozza et. al. [23]-[24]. The HTO test involved a constant power dissipation of 6W/mm at V_{DS}=25V and I_{DS}=240 mA/mm. at various ambient temperatures of 100°C, 120°C and 140°C, leading to estimated channel temperatures of 204°C, 232°C and 260°C. Before the HTO test, the noise spectra for the fractional fluctuations in the channel resistance and current are 1/f spectra with remnants of some generation-recombination (GR) bumps from traps generating lorentzian spectral contributions. After the HTO test, the samples just show a 3 orders of magnitude larger, perfect fundamental 1/f, noise spectrum. The channel resistance increases about 5 times during the HTO test. The damage is likely to be concentrated on certain regions of the large device width of 200-400 μm. In those regions the channel is likely to be damaged mainly between the gate and drain, that is sub-threshold, has the lowest conductance, the highest hot electron and phonon thermal dissipation, and suffers the highest structural degradation. As a result of this degradation a 2 orders of magnitude increase of the 1/f fractional channel resistance fluctuations noise power spectrum could be initiated by the H-instability and caused only in part by the reduction of the effective channel width due to non-uniformity. It should induce bulk GaN current transport contribution (base leakage) with a giant piezoelectric quantum 1/f coefficient α_{piezo} of the order of unity that is applicable also to the gate leakage current, causing in large part the huge 1/f noise increase found after the HTO test. Note that other tests, such as the high temperature reverse bias (HTRB), and high forward gate current (HFGC) tests, only showed [23]-[24] variations of the order of 10% in resistance and noise. In the larger scheme, we might also mention that Nikonov (INTEL) and Zhirnov have shown [25] that electronic 1/f noise sets the limit of Moore's law.

10. CONCLUSIONS

The theoretical values of the noise parameter α for both ELO-GaN and Si doped n^+-GaN:Si/Al_2O_3 epilayers were estimated with the strain dependence dimensionless piezopolaron coupling coefficient *g* based on the strain dependence transversal acoustic phonon velocity. The experimentally obtained α for both ELO-GaN and n^+-GaN:Si samples correlate well with the theoretically estimated α. The energy-momentum *E*(*P*) relationships given in Table 8.1 allows us to estimate the polaron self-energy as well as the polaron mass. Since the piezopolaron coupling coefficients *g* for both ELO-GaN and n^+-GaN:Si are greater than 3π/4, the strong coupling theory will be appropriate here. For ELO-GaN, the polaron self-energy estimated was -1.91meV and the polaron mass was m* = 0.282m. Since the estimated polaron mass is heavier than the effective electron mass which is given as m* = 0.2m, the estimation of the strain dependence piezopolaron coupling coefficient for ELO-GaN is justifiable. In conclusion, the theoretically estimated noise parameter α based on coherent piezoelectric quantum 1/*f* noise agrees with the experimentally obtained results for ELO-GaN. This result indicates that piezoelectric effect is indeed relevant both for the transport process in ELO-GaN and for its fluctuations. For QED our calculation in FETs and HFETs showed that only an effective width, about equal to the device length L should be used in the calculation of the coherence parameter s in this special case, because of the weak dependence of the natural logarithm on its argument.

For lattice-dynamical (piezoelectric Q1/f noise based on TA infraquanta) the weight parameter, or coherence parameter, s' has the same role as the parameter s had for the usual (QED) Q1/f noise. It gives a rough interpolation approximation of the piezoelectric Q1/f coefficient by combining the conventional and coherent parts in a weighted sum

$$\boldsymbol{\alpha}_H = fNS_{\delta j/j}(f) = [1/(1+s')](4g/3\pi)(\boldsymbol{\Delta k}/k_s)^2 + [s'/(1+s')]\boldsymbol{\{}2g/\pi[1-(<\mathbf{k}>/k_s)^2]\boldsymbol{\}}, \tag{10.1}$$

where v_s is the speed of sound, $<\mathbf{k}>=m^*<\mathbf{v}>/\hbar$, $|<\mathbf{v}>| = u$ is the drift velocity of the carriers, $\mathbf{\Delta k}/k_s=\mathbf{\Delta v}/v_s$, and

$$k_s = m^*v_s/\hbar; \quad s' = (gN'\hbar/\pi m^*v_s)(v_s/u)^3F(u/v_s). \tag{10.2}$$

Here N' is the number of carriers per unit length of the sample in the direction of current flow, and

$$F(u/ v_s) = 2(u/ v_s)\{1 + 1/[1 - (u/ v_s)^2]\} + \ln[1-(u/ v_s)]^2 - \ln[1+(u/ v_s)]^2. \tag{10.3}$$

The coherence parameter s' was calculated in the same way in which s was introduced in the QED case for the connection between coherent and conventional Q1/f). It is the ratio of the coherent piezoelectric energy of the drift motion divided by the additional kinetic energy of the carriers introduced by the drift velocity u.

Finally the piezoelectric coherence parameter s' for the special case of FETs and HFETs witw>>L_{DS}>t

$$s' == (gN'\hbar/m^*v_s)(v_s/u)^3F(u/v_s)t/12w, \tag{10.4}$$

was derived here for the first time. It can easily be generalized to the case of several sheets with different piezoelectric properties in the above integral over t'. We notice that the coherent contribution is strongly reduced due to the smallness of t as compared to w. The conventional Q1/f contribution is inherently smaller, since for $v<v_s$ it contains the additional small factor $(\Delta v/v_s)^2$. We conclude that the strong decrease in s' present in thin samples or channels (small t) could explain the startling reduction in the observed piezoelectric 1/f noise in modern HFETs.

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